

A certified 67.32001% lower-bound candidate for simple zeros of the Riemann zeta function on the critical line

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Abstract

Let $N(T, 2T)$ count nontrivial zeros of the Riemann zeta function with $T < \gamma \leq 2T$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line. We give machine-certified finite data which, through the arbitrary-window Gabor interface of Anthropic and the stability refinement of `ainta/zeta-simple-zeros`, imply

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.6732001170127618568 \dots > \frac{1683}{2500}.$$

This improves the preceding 0.673137630699... construction. The new finite ingredient is a re-tuned position-weighted seven-point gap inequality, proved by exhaustive interval subdivision, combined with the existing re-optimized window and sharp square-root Gram-defect profile. The finite inequality, window estimates, span capacities, and final arithmetic are certified with Arb. The new assembly has not been formalized end to end in Lean, so the result is presented as a research-draft record candidate pending independent mathematical review.

Two independent same-day repositories, located after the finite certificate and initial draft were complete, report candidates 0.673192911473... and 0.673195198901... [4, 5]. The present value is numerically higher than the latter by 0.0000049181112412813... in proportion. Neither artifact was an input to this construction, and the numerical comparison establishes neither accepted priority nor independent validity.

1 Statement and proof boundary

Theorem 1 (certified reduction). *Assume the arbitrary-admissible-window asymptotic interface stated in [1, §7.1] and the stability and shifted-block interfaces of [2], with their stated uniformity. Then*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.6732001170127618568 \dots > \frac{1683}{2500}.$$

The hypothesis in Theorem 1 is a proof boundary, not a numerical assumption. The cited works supply those analytic statements; this repository certifies every new finite assertion needed

*This work was developed in a ChatGPT Work conversation between the author and OpenAI Codex. Codex performed the numerical search, implementation, interval-certificate construction, and much of the initial drafting under the author's direction. The result builds on the 67.31376% construction credited in Appendix B; full AI and upstream provenance is recorded there and in `docs/provenance.md`. This contributor draft does not presume the authorship arrangement of any merged or published version.

to instantiate them. Unlike the headline theorems in the Anthropic Lean artifact, however, this particular window, the stability refinement, and the square-root assembly have not yet been joined in one formal proof.

The framework of [1, 2] uses Weil’s explicit formula on a Gabor family of windowed test functions. It produces a real symmetric matrix $\tilde{G}$ whose zero side splits into a positive-semidefinite part from on-line zeros and signature- $(1, 1)$ blocks from off-line pairs. Montgomery’s first and second moments evaluate $\text{tr } \tilde{G}$ and $\|\tilde{G}\|_F^2$. The refinement of [2] retains a stability defect $\text{tr } \Psi(M)$, where M is the Gram matrix of the simple-zero atoms and

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases}$$

We import the explicit-formula and trace asymptotics, the tail estimates, and the compact-uniform Gram-entry asymptotic from [1]; from [2] we import the stability rank–trace inequality, convex pinching, and shifted-block averaging.

2 The window

For an even profile $v \geq 0$ on $[-\frac{1}{2}, \frac{1}{2}]$ (the square of the taper, before the standard boundary ramp), define

$$c_1(v) = \frac{(\int v)^2}{\int v^2 + \iint |s-s'|v(s)v(s')dsds'}, \quad H(v) = 2 - \frac{1}{c_1(v)},$$

and the normalized overlap kernel

$$k_v(x) = \frac{K_v(x)}{K_v(0)}, \quad K_v(x) = \int_{-1/2}^{1/2} v(t) \cos(2\pi xt) dt.$$

The Montgomery–Taylor profile maximizes H alone. The defect term below responds at first order to a profile perturbation while the loss in H is second order, creating a small net improvement.

We use $v(s) = \sum_{j=0}^6 c_j \cos(\omega_j s)$ with

$$\omega = (\sqrt{2}, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi, 12\pi)$$

and exact rational coefficients

$$c = \frac{1}{10^9}(10^9, 3322500, -7609135, 1190194, -731476, -1680572, 1141360).$$

Proposition 2 (certified). *On $[-\frac{1}{2}, \frac{1}{2}]$,*

$$\frac{3}{4} \leq v \leq 1, \quad \frac{v'(s)}{s} < -0.7763 \quad (0 < s \leq \tfrac{1}{2}),$$

and

$$H(v) = 0.6724570414145442887\dots \geq H_{\text{cert}} := \frac{3362285207}{5000000000}.$$

In particular, v is nonincreasing on $[0, \frac{1}{2}]$.

Proof. Arb interval evaluation on an 8192-cell subdivision proves the range and monotonicity bounds. At the removable zero,

$$\frac{v'(s)}{s} = - \sum_j c_j \omega_j^2 \operatorname{sinc}(\omega_j s), \quad \operatorname{sinc}(z) = \frac{\sin z}{z}.$$

The functional is evaluated from the closed forms

$$\int v = \sum_j c_j \operatorname{sinc}(\omega_j/2), \quad \int \cos(as) \cos(bs) ds = \frac{1}{2} \left[\operatorname{sinc} \frac{a-b}{2} + \operatorname{sinc} \frac{a+b}{2} \right],$$

and the corresponding exact expression for $\iint |s-t| \cos(as) \cos(bt) ds dt$. The recorded Arb enclosure is `certificates/fast-parts.txt`. $\square$

Because v is positive and nonincreasing, $\sqrt{v}$ followed by the fixed outward-monotone boundary ramp of [1, §2.2] has total variation at most 2; the same holds for its square. Positivity $v \geq 3/4$, the finite trigonometric expansion, and the fixed ramp give the required first- and second-derivative bounds with a profile-dependent constant. The fourth moment condition also holds eventually: away from the $O(1)$ ramp, $\int v^2 \geq 9/16 > 1/2$. Thus the cited arbitrary-window interface gives, for $S = N_0^s(T, 2T)$ and $N = N(T, 2T)$,

$$S \geq H(v)N + \operatorname{tr} \Psi(M) - o(N). \quad (1)$$

Writing an explicit Lean `AdmWindow` instance remains future work.

3 A sharp square-root tail for the block defect

Write $w = k_v^2$. For a positive-semidefinite block G set

$$E = 2 \sum_{i < j} |G_{ij}|^2, \quad \Delta = \operatorname{tr} \Psi(G).$$

Lemma 3. $\Delta \geq h(E)$, where

$$h(E) = \begin{cases} E, & 0 \leq E \leq 1, \\ 2\sqrt{E} - 1, & E \geq 1. \end{cases}$$

The profile is sharp.

Proof. Use $\Psi(t) = (t-1)^2 - (t-2)_+^2$. For the eigenvalues $\lambda_i \geq 0$ put $X = \sum_i (\lambda_i - 2)_+^2$. Then

$$E \leq \sum_i (\lambda_i - 1)^2 = \Delta + X.$$

If $X = 0$, then $\Delta \geq E$. If $X > 0$, each $\lambda_i > 2$ contributes $1 + 2z_i$ to Δ , where $z_i = (\lambda_i - 2)_+$. Consequently $\sum_i z_i \leq (\Delta - 1)/2$ and $X \leq (\sum_i z_i)^2 \leq (\Delta - 1)^2/4$. Thus $E \leq (\Delta + 1)^2/4$, so $\Delta \geq 2\sqrt{E} - 1$; moreover $\Delta \geq 1 \geq E$ when $E \leq 1$. Sharpness follows from one eigenvalue $a > 2$ and $n - 1$ equal eigenvalues $(n - a)/(n - 1)$, conjugated to constant diagonal, and then letting $n \rightarrow \infty$. $\square$

Since h is increasing and concave with $h(0) = 0$, for every $A > 0$ and $\eta = h(A)/A$,

$$h(E) \geq \eta \min(E, A). \quad (2)$$

4 The certified seven-point inequality

For gaps $g_1, \dots, g_6 \geq 0$ between seven consecutive simple zeros put $y_0 = 0$ and $y_j = g_1 + \dots + g_j$. Define

$$F(g) = \frac{1}{2736} \sum_{i=1}^6 g_i + \sum_{0 \leq i < j \leq 6} a_{ij} w(y_j - y_i). \quad (3)$$

The exact numerators $10^6 a_{ij}$ are

$i \backslash j$	1	2	3	4	5	6
0	236484	535224	969656	1000000	1000000	2000000
1		381281	464776	30344	0	1000000
2			382235	0	30344	1000000
3				382235	464776	969656
4					381281	535224
5						236484

and are reflection symmetric. Direct rational summation gives, for every span $1 \leq r \leq 6$,

$$\sum_{i=0}^{6-r} a_{i,i+r} = 2. \quad (4)$$

Proposition 4 (certified). *For every $g \in [0, \infty)^6$,*

$$F(g) \geq \frac{891}{200000} = 0.004455.$$

Proof. An exhaustive interval subdivision with grid $1/4000$ visited 2,168,370 boxes and certified every one, reaching maximum depth 50. Arb constructs the kernel and derivative tables; all conversions to binary64 are rounded outward. The branch-and-bound proof uses pressure, direct interval, and convex-tangent lower bounds. Near removable singularities in the sinc derivatives, it switches to exact-sign rational branch tests and alternating series with explicit tail bounds. The complete transcript and table hashes are in `certificates/retuned-p2736-grid4000.txt`. The observed floating-point minimum $0.0044623605595\dots$ is used only as a discovery diagnostic; it is about 0.16% above the certified target. $\square$

5 Deduction of Theorem 1

Set

$$q = 6, \quad m = 272, \quad \varepsilon = \frac{891}{200000}, \quad p = \frac{1}{2736}, \quad B_p = qp = \frac{1}{456}.$$

Step 1 (window sum). For a block B of m consecutive simple zeros, sum (3) over its $m - q$ consecutive seven-point windows. A pair spanning r gaps receives total weight at most 2 by (4), and each gap appears in at most q windows. With $E_B = 2 \sum_{i < j} w(y_j - y_i)$ and $A := \varepsilon(m - q) = 1.18503$, this gives

$$E_B + B_p \text{span}(B) \geq A. \quad (5)$$

Step 2 (bounded-span case split). Put

$$R = h(A) = 2\sqrt{A} - 1 = 1.177181664446\dots, \quad \eta = \frac{R}{A} = 0.993377099690\dots$$

If $B_p \text{span}(B) \geq A$, then $\eta B_p \text{span}(B) \geq R$, so the pressure term alone wins. Otherwise $\text{span}(B) < A/B_p$, a fixed constant. The compact-uniform Gram-kernel asymptotic therefore applies simultaneously to all pairs in the fixed-size block, and the Gram off-diagonal energy equals $E_B - o(1)$ uniformly. Equations (2) and (5) then yield

$$\text{tr } \Psi(G_B) + \eta B_p \text{span}(B) \geq R - o(1). \quad (6)$$

Indeed, if $E_B \geq A$ the defect supplies $R - o(1)$; otherwise the two terms supply at least $\eta(E_B + B_p \text{span}(B)) - o(1)$.

Step 3 (averaging). Remove the standard endpoint strips of normalized width L^2 in the mesoscopic notation of [2]; their contribution is $o(N)$. Convex spectral pinching gives $\text{tr } \Psi(M) \geq \sum_B \text{tr } \Psi(G_B)$ for disjoint blocks. Average (6) over the m shifted block partitions. Every interior gap is charged in at most $m - 1$ offsets, the total normalized span is $N + o(N)$, and leftover endpoint blocks contain only $O(m)$ points per offset. Therefore

$$\text{tr } \Psi(M) \geq \frac{R}{m} S - \eta B_p \frac{m-1}{m} N - o(N). \quad (7)$$

Step 4 (assembly). Insert (7) into (1), use $H(v) \geq H_{\text{cert}}$, and rearrange:

$$\frac{S}{N} \geq \frac{272H_{\text{cert}} - \eta(1/456)271}{272 - R} - o(1) = 0.6732001170127618568 \dots > \frac{1683}{2500}.$$

The final Arb enclosure has width below 10^{-50} and is recorded in `certificates/fast-parts.txt`. Replacing the data by the preceding 67.313763% design reproduces its value, while the separate gate reproduces the [2] seven-point certificate.

Remark 5. This optimization does not offer a route to 100%. The obstruction for certificates using only the bandwidth-one data in the Anthropic Lean artifact is $0.68182868746 \dots$, while the subsequent Campaign 2 analysis gives the narrower pure pair-energy class a ceiling near 0.674826. A 100% liminf result would require qualitatively stronger information; even then it would establish a density-one statement, not rule out every exceptional zero.

A Verification summary

Claim	Method	Record
$3/4 \leq v \leq 1$ and monotonicity	8192-cell Arb evaluation	<code>fast-parts.txt</code>
$H(v) \geq 3362285207/(5 \cdot 10^9)$	closed forms in Arb	<code>fast-parts.txt</code>
$F \geq 891/200000$	2,168,370-box subdivision	<code>retuned-p2736-grid4000.txt</code>
final arithmetic	Arb, 256-bit	<code>fast-parts.txt</code>
gate: ainta seven-point	707,797 boxes	<code>gate-repro-seven.txt</code>

Reproduction requires Python 3.10 or newer. After `pip install -e .`, run

`zeta-673200-verify all.`

The dependency `python-flint==0.9.0` is pinned, and the main certificate records hashes of its outward-rounded tables. The verifier fails closed if any terminal grid cell remains unresolved.

B Provenance

This repository revision starts from commit `e0a2266` of [3]. That work supplied the perturbed window, square-root profile, generalized verifier, and the preceding 67.313763% design; its multi-model contribution record is preserved in `docs/provenance.md`. The present follow-up was initiated and directed by Vivaswat Ojha in a ChatGPT Work conversation with OpenAI Codex. It keeps the window fixed and retunes the pressure, exact pair weights, and block length. Codex carried out the numerical search, implementation, certificate construction, verifier hardening, and much of the initial drafting; Ojha is the human author and takes responsibility for submitting this contributor draft. The upstream maintainer is invited to determine how any merged or published version should be organized and attributed, without disturbing this contribution record. The revision also makes the bounded-span case split and endpoint bookkeeping explicit. Numerical optimization was used only to discover a rational candidate. The exhaustive interval certificate is the decision procedure.

References

- [1] Anthropic, *More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, August 2026. Paper and Lean 4 artifact, <https://github.com/anthropics/zeta-23-lean>.
- [2] `ainta/zeta-simple-zeros`, *A 67.30085% lower bound for simple zeros of the Riemann zeta function*, 2026. Research repository, <https://github.com/ainta/zeta-simple-zeros>.
- [3] `trmdy/zeta-simple-zeros-673137`, *A 67.31376% lower-bound candidate for simple zeros of the Riemann zeta function*, 2026. Research repository, <https://github.com/trmdy/zeta-simple-zeros-673137>.
- [4] `tawanerguo-cn/zeta-simple-zeros`, *Certified unconditional 67.3192911473% lower bound for simple zeros of the Riemann zeta function*, 2026. Research archive, <https://github.com/tawanerguo-cn/zeta-simple-zeros>.
- [5] N. Pipitone, *A 67.3195% lower bound for simple zeros of the Riemann zeta function*, 2026. Research repository, <https://github.com/npip99/zeta-zeros>.
- [6] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, 1973; H. L. Montgomery and R. M. Taylor, 1975.
- [7] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya, C. L. Turnage-Butterbaugh, *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta-function*, *Acta Arith.* 214 (2024).